

Yuanhao Bai

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EDUCATION

Central South University, Changsha, China

Sep 2023 – Jun 2027

B.Eng. in Safety Engineering, Expected Jun 2027

Upper-Level Coursework Average: 89.09/100 | Rank: 12/71 in Major (Top 17%)

Relevant Background: Machine Learning; Deep Learning; Python Programming; Probability and Mathematical Statistics; Linear Algebra; Circuit Principles; Analog and Digital Circuits; Advanced Mathematics.

Research Interests: AI Systems and Infrastructure; Multi-Agent Reinforcement Learning and Embodied AI; World Models and Multimodal Perception.

PUBLICATIONS & PATENTS

G. Yao*, Y. Bai*, and T. Chang, “AutoRole-Tiny: Single-Genome Role Differentiation for Collective Inference on MCU Sensor Swarms,” manuscript in preparation for submission to ACM/IEEE SenSys 2027.

* These authors contributed equally.

K. Wang, H. Zheng, Y. Bai, et al., “SAtom: Swarms as Tools for Object Manipulation,” manuscript in preparation for submission to *IEEE Robotics and Automation Letters (RA-L)*.

Invention patent application No. 202610514577.7 — First-listed Inventor. FireGuard multimodal fire-risk early warning and intervention method/system; accepted by CNIPA.

Invention patent application No. 202611195180.2 — Co-inventor (second-listed). Multimodal driver fatigue and distraction detection method/system; accepted by CNIPA; fast pre-examination passed.

RESEARCH EXPERIENCE

ATOMIC Lab, IoT Thrust, Information Hub, HKUST(GZ)

Jul 2026 – Present

Research Assistant — TinyML Systems and AI Infrastructure

Advisor: Tengfei Chang

- **Co-developed AutoRole-Tiny**, a decentralized TinyML system that executes recurrent multi-agent inference across homogeneous nRF52840 sensor nodes using local observations, fixed-size message buffers, and permutation-invariant aggregation.
- Built its end-to-end **PyTorch-to-MCU INT8 inference infrastructure**, covering fixed Q0.7 QAT and model export, bit-exact Python/C++ validation, static tensor-lifetime and activation-memory planning, **CMSIS-NN kernel optimization**, and profiling of Flash, SRAM, latency, and PPK2 energy.

Huawei–University of Macau Joint Research Project

Sep 2026 – Present

State Key Laboratory of Internet of Things for Smart City, University of Macau

AI Infrastructure Research Contributor — Efficient NPU Inference Runtime

- Investigating **INT8 vision-model deployment** on NXP i.MX93 / Arm Ethos-U65 for resource-constrained smart-glasses workloads, including **onboard MobileNet validation** and **EfficientDet-Lite0 multi-partition execution characterization**.
- Established a **Docker-to-EVK experimental workflow** with **model-integrity validation and temperature monitoring**, and am developing **latency, thermal, and power profiling** under sustained inference.

AIFMWS Lab, Central South University

Mar 2025 – Jun 2026

Project Lead — FireGuard, Undergraduate Open Laboratory Project

Advisor: Prof. Zhi Li

- Led a **national-level undergraduate innovation project** developing a **Jetson Nano + ESP32 multimodal fire-risk detection and intervention system**; built an **HIL testbed** exposing real-time MAC-layer states for distributed decision-making under bandwidth constraints; received an **Excellent completion rating** and filed a **Chinese invention patent application** as **first-listed inventor**.

SELECTED PROJECTS

SAtom: Swarms as Tools for Object Manipulation Jul 2026 – Present

Research Assistant, ATOMIC Lab, HKUST(GZ) | Target: IEEE RA-L

- Conducted **physical-system evaluation and demo validation** for a **foundation-model-driven swarm manipulation framework** based on collective virtual end-effectors, contributing to **120 physical trials with 89.2% success and 10.4 mm mean placement error**.

RK3568 Edge Vision System for Oriented Fastener Inspection Jul 2026

Embedded AI Deployment and Validation Contributor — 9th National Embedded Chip and System Design Competition (Southern Regional Round)

- Contributed to a **13-class YOLOv8s-OBB system with INT8 RKNN inference on QSM368ZP-WF/RK3568**, achieving **99.1% mAP@50** and **94.7% mAP@50–95**; **built reproducible calibration, conversion, board-test, and hard-case analysis tooling**.

LLM-Augmented Multi-Agent Emergency Evacuation Simulation Mar 2026 – May 2026

Project Lead — First Prize, 10th Central South University Virtual Simulation Design Competition

- Built a **Unity/Tuanjie-based emergency simulation platform** integrating thermal-runaway progression, emergency intervention, and pedestrian evacuation in EV charging garages.
- Implemented an **LLM-augmented multi-agent decision module** that enables pedestrians to choose actions from local hazards and drive Unity navigation and state transitions.

Multimodal Driver Fatigue and Distraction Monitoring Jan 2026 – Present

Project Lead / Co-inventor (second-listed) Advisor: Longjun Dong

- Designed a **multimodal driver-state inference framework** that fuses visual, audio/speech, and smoking-related gas/smoke sensing through **temporal verification and reliability-aware risk aggregation**.
- Developed an **end-to-end perception-to-warning pipeline** with graded decision logic for robust fatigue and distraction monitoring under heterogeneous sensor uncertainty; contributed the algorithm framework to a Chinese invention patent application.

SELECTED HONORS AND AWARDS

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| [1–3] | National First Prize , 12th National College Student Safety Science and Engineering Practice and Innovation Works Competition — Team Lead ; also National Third Prize (2025) and National Second Prize (2024) , both as Team Lead | 2026 |
| [4] | National-level Undergraduate Innovation and Entrepreneurship Training Program — Excellent Completion , Project Lead | 2026 |
| [5] | First Prize , 10th Central South University Virtual Simulation Design Competition — Project Lead | 2026 |
| [6] | Excellent Paper Award and Excellent Case Award , CSU Undergraduate Innovation Annual Conference | 2026 |

TECHNICAL SKILLS

Programming and AI: Python, C/C++, Git/GitHub; PyTorch, OpenCV, recurrent and multi-agent models, QAT/INT8 inference, and multimodal fusion.

Embedded deployment and profiling: nRF52840/Zephyr, BLE, ESP32, Jetson Nano, RK3568/RKNN, and CMSIS-NN; Python/C++/MCU consistency, latency, Flash/RAM, and PPK2 power/energy profiling.